

Comprehensive Sovereign Map Model

System Reference: MAP-SOVEREIGN-V1

This document defines the machine-readable, fillable-form architecture for a universal sovereign nation dataset. It serves as a structural map to ingest, validate, and bridge core geopolitical data points—including ISO standards, geographic coordinates, and time zones—across 249 global entities.

1. Data Schema Architecture (JSON Validation Template)

JSON

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "SovereignMapEntity",
  "type": "object",
  "properties": {
    "entity_metadata": {
      "type": "object",
      "properties": {
        "common_name": {
          "type": "string",
          "examples": "{{COMMON_NAME}}"
        },
        "formal_name": {
          "type": "string",
          "examples": "{{FORMAL_NAME}}"
        },
        "wikidata_qid": {
          "type": "string",
          "pattern": "^Q[0-9]+$",
          "examples": "{{WIKIDATA_QID}}"
        },
        "niem_compliance_version": {
          "type": "string",
          "const": "6.0"
        }
      }
    },
    "required": "common_name" "wikidata_qid",
    "iso_codes": {
      "type": "object",
      "properties": {
        "alpha_2": {
          "type": "string",
          "pattern": "^[A-Z]{2}$",
          "examples": "{{ISO_A2}}"
        },
        "alpha_3": {
          "type": "string",
          "pattern": "^[A-Z]{3}$",
          "examples": "{{ISO_A3}}"
        },
        "numeric": {
          "type": "string",
          "pattern": "^[0-9]{3}$",
          "examples": "{{ISO_NUM}}"
        }
      },
      "required": "alpha_2" "alpha_3" "numeric",
      "geographic_spatial_anchor": {
        "type": "object"
      }
    }
  }
}
```

```

    "properties"
    "centroid_coordinates"
    "type" "object"
    "properties"
      "latitude" "type" "number" "minimum" -90.0 "maximum" 90.0 "examples" 0.0000
      "longitude" "type" "number" "minimum" -180.0 "maximum" 180.0 "examples" 0.0000
    ]
    "required" "latitude" "longitude"
  ]
  "bounding_box"
  "type" "array"
  "minItems" 4
  "maxItems" 4
  "items" "type" "number"
  "description" "[Min Longitude, Min Latitude, Max Longitude, Max Latitude]"
]
"required" "centroid_coordinates"
]
"temporal_zones"
"type" "array"
"items"
"type" "object"
"properties"
  "iana_identifier" "type" "string" "examples" "{{IANA_ZONE_ID}}"
  "utc_offset_standard" "type" "string" "pattern" "^[+-][0-9]{2}:[0-9]{2}$"
]
"required" "iana_identifier" "utc_offset_standard"
]
]
]
"required" "entity_metadata" "iso_codes" "geographic_spatial_anchor" "temporal_zones"
]

```

2. Functional Ingestion Form (Fillable Map Template)

Python

```
#!/usr/bin/env python3
"""
Module: map_ingestion_engine.py
Version: 3.1.2
Description: General/Form version for mapping global geopolitical entities
to NIEM 6.0 and Wikidata tracking layers.
"""
```

```
import sys
from typing import Dict, Any, List
```

```
class SovereignMapNode:
    def __init__(self):
        # Fillable slots indicating accepted data types
        self.common_name: str = "{{STR:COMMON_NAME}}"
        self.formal_name: str = "{{STR:FORMAL_NAME}}"
        self.qid: str = "{{STR:WIKIDATA_QID}}"

        self.iso_a2: str = "{{STR_EXACT_2:ISO_ALPHA_2}}"
        self.iso_a3: str = "{{STR_EXACT_3:ISO_ALPHA_3}}"
        self.iso_num: str = "{{STR_EXACT_3_DIGIT:ISO_NUMERIC}}"

        self.latitude: float = 0.0000 # {{FLOAT:CENTROID_LAT}}
        self.longitude: float = 0.0000 # {{FLOAT:CENTROID_LON}}

        self.timezones: List[Dict[str, str]] = [] # {{LIST_DICT:IANA_ZONES}}
```

```
def export_map(self) -> Dict[str, Any]:
    """Compiles the runtime form into a structured, machine-readable dictionary mapping."""
    return {
        "entity_metadata": {
            "common_name": self.common_name,
            "formal_name": self.formal_name,
            "wikidata_qid": self.qid,
            "niem_compliance_version": "6.0"
        },
        "iso_codes": {
            "alpha_2": self.iso_a2,
            "alpha_3": self.iso_a3,
            "numeric": self.iso_num
        },
        "geographic_spatial_anchor": {
            "centroid_coordinates": {
                "latitude": self.latitude,
```

```

        "longitude" self.longitude
    ]

    "temporal_zones" self.timezones

def validate_node(self) -> bool:
    """Executes strict structural and validation rules against standard spatial-temporal bounds."""
    if not 90.0 <= self.latitude <= 90.0:
        return False
    if not 180.0 <= self.longitude <= 180.0:
        return False
    if len(self.iso_a2) != 2 or len(self.iso_a3) != 3:
        return False
    return True

if __name__ == "__main__":
    # Execution structural verification example
    node = SovereignMapNode()
    # Dynamic insertion block for template automation pipelines
    print(json.dumps(node.export_map(), indent=2))

```

3. Structural Translation Layer (NIEM 6.0 <=> Wikidata Q-ID)

This crosswalk provides structural rules for converting arbitrary spatial tracking objects between international frameworks:

Source Property (Wikidata)	Schema Mapping Pathway	Target Element (NIEM 6.0)	Data Constraint
P297 (ISO 3166-1 alpha-2)	iso_codes.alpha_2	nc:CountryAlpha2Code	2-Character Upper-case String
P298 (ISO 3166-1 alpha-3)	iso_codes.alpha_3	nc:CountryAlpha3Code	3-Character Upper-case String
P299 (ISO 3166-1 numeric)	iso_codes.numeric	nc:CountryNumericCode	3-Digit Zero-padded String
P625 (Coordinate Location)	geographic_spatial_anchor.centroid_coordinates	nc:LocationGeographicPoint	Floating point pair

	ordinates		
P421 (Located in Time Zone)	temporal_zones[].iana_identifier	nc:LocationTimeZoneName	Valid IANA ID text